

Hints & Solutions

#MathBoleTohMathonGo

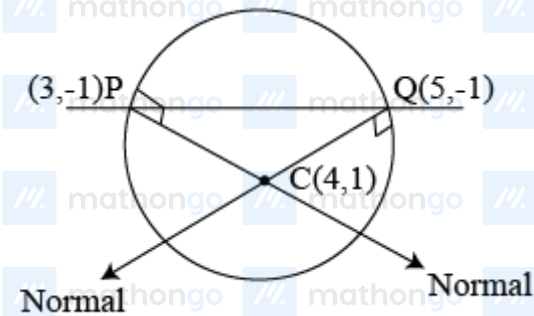
Q1 (1) - Tangent and Normal

When, $y = -1$

$$x^2 + 1 - 8x + 2 + 12 = 0$$

$$x^2 - 8x + 15 = 0$$

$$\Rightarrow x = 3, 5$$

 $\therefore P(3, -1)$ and $Q(5, -1)$ $\Rightarrow$ Equation of normal at P and Q are

$$\frac{y-1}{x-4} = \frac{-1-1}{5-4}$$

$$\Rightarrow y - 1 = -2x + 8$$

$$\text{i.e. } y - 9 = 0$$

$$\text{and } \frac{y-1}{x-4} = \frac{-1-1}{3-4}$$

$$\Rightarrow y - 1 = 2x - 8$$

$$\Rightarrow 2x - y - 7 = 0$$

Q2 (3) - Tangent and Normal

The given equation of line is

$$x + y = 2$$

It can be rewritten as

$$y = -x + 2$$

$$\text{Slope} = -1$$

If the line touches the circle, then (length of perpendicular from the centre $(c, 0)$ = radius)

$$\left| \frac{(1)(c) - 0 - 2}{\sqrt{1+1}} \right| = c$$

$$\Rightarrow \left| \frac{c-2}{\sqrt{2}} \right| = c$$

$$\Rightarrow c - 2 = \pm c\sqrt{2}$$

$$\Rightarrow c(1 \mp \sqrt{2}) = 2$$

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$$\Rightarrow c = \frac{2}{1 \mp \sqrt{2}}$$

$$\Rightarrow c = \frac{1}{1+\sqrt{2}}, \frac{2}{1-\sqrt{2}}$$

$$\Rightarrow c = 2(\sqrt{2} - 1), -2(\sqrt{2} + 1)$$

Since, (1,0) lies inside the circle. $\Rightarrow$ Distance between the point (1,0) and centre will be less than radius.

$$\therefore \sqrt{(c-1)^2 + (0-0)^2} < c$$

$$\Rightarrow \sqrt{(c-1)^2} < c$$

$$\Rightarrow |c-1| < c$$

$$\Rightarrow -c+1 < c \text{ or } c-1 < c$$

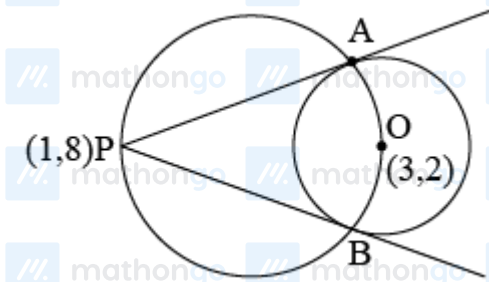
$$\Rightarrow 2c > 1-1 > 0$$

$$\Rightarrow c > 1$$

Hence, required value of $c = 2(\sqrt{2} - 1)$

Q3 (2) - Tangent and Normal

For required circle $P(1, 8)$ and $O(3, 2)$ will be the end point of its diameter.



$$\therefore (x-1)(x-3) + (y-8)(y-2) = 0$$

$$\Rightarrow x^2 + y^2 - 4x - 10y + 19 = 0$$

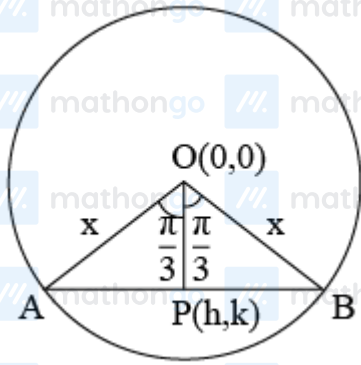
Q4 (3) - Tangent and Normal

Let $P(h, k)$ be the mid-point of the chord subtending the angle $\frac{2\pi}{3}$ at the centre.

In $\triangle OPA$, we have

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$$\cos \frac{\pi}{3} = \frac{OP}{OA}$$

$$\Rightarrow 2OP = OA$$

$$\Rightarrow 4OP^2 = OA^2$$

$$\Rightarrow 4(h^2 + k^2) = 9$$

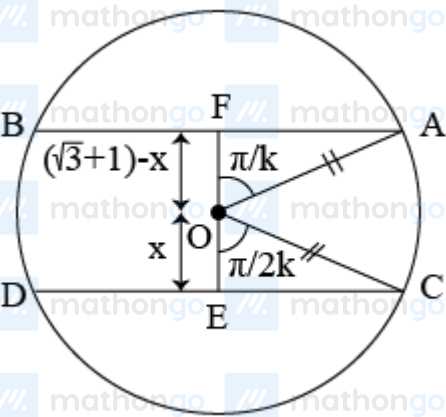
$$\therefore \text{Locus of } (h, k) \text{ is } x^2 + y^2 = \frac{9}{4}$$

Q5 (2) - Tangent and Normal

$$\text{Let } Q = \frac{\pi}{2k}$$

$$\therefore \cos \theta = \frac{x}{2}$$

$$\text{and } \cos 2\theta = \frac{(\sqrt{3}+1)-x}{2}$$



$$\Rightarrow \cos 2\theta = \frac{\sqrt{3}+1}{2} - \cos \theta$$

$$\Rightarrow \cos 2\theta + \cos \theta = \frac{\sqrt{3}}{2} + \frac{1}{2}$$

$$\therefore \theta = 30^\circ = \frac{\pi}{6}$$

$$\Rightarrow \frac{\pi}{2k} = \frac{\pi}{6} \Rightarrow k = 3$$

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Q6 (4) - Angle Between Circles and Radical Axis

The equation of the circle having centre (2,-2) is

$$S_1 \equiv (x - 2)^2 + (y + 2)^2 = 0$$

$$\Rightarrow S_1 \equiv x^2 + y^2 - 4x + 4y + 8 = 0 \dots(i)$$

$$S_2 \equiv x^2 + y^2 - 2x + 3y + \frac{1}{3} = 0 \dots(ii)$$

The radical axis of two circles, $S_1 = 0$ and $S_2 = 0$ is given by $S_1 - S_2 = 0$

$$(-4x + 4y + 8) - \left(-2x + 3y + \frac{1}{3}\right) = 0$$

$$\Rightarrow -2x + y + \frac{23}{3} = 0$$

$$\Rightarrow -6x + 3y + 23 = 0$$

$$\Rightarrow 6x - 3y - 23 = 0$$

Q7 (3) - Angle Between Circles and Radical Axis

The given equations of the circles are

$$(x - p)^2 + y^2 = p^2$$

$$\Rightarrow x^2 + y^2 - 2px = 0 \dots(i)$$

$$x^2 + (y - q)^2 = q^2$$

$$\Rightarrow x^2 + y^2 - 2qy = 0 \dots(ii)$$

$$\text{and } x^2 + y^2 = p^2 + q^2 \dots(iii)$$

From Eqs. (i) and (iii), we get

$$2px = p^2 + q^2$$

$$\Rightarrow x = \frac{p^2 + q^2}{2p}$$

From Eqs. (ii) and (iii), we get

$$2qy = p^2 + q^2$$

$$\Rightarrow y = \frac{p^2 + q^2}{2q}$$

$$\therefore \text{Coordinates of centre} \left(\frac{p^2 + q^2}{2p}, \frac{p^2 + q^2}{2q} \right)$$

Q8 (3) - Angle Between Circles and Radical Axis

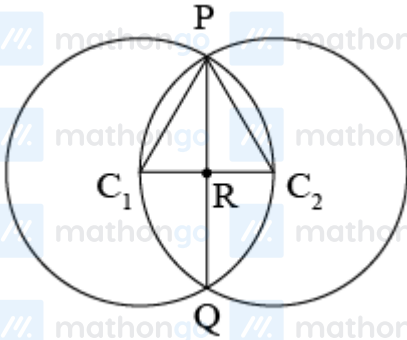
Both the circles have same radius

$$\text{i.e. } C_1P = C_2P = r$$

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where, C_1 and C_2 are their centres.



PQ is the common chord of both the circles and $C_1C_2 = r$. In ΔC_1PR ,

$$C_1P = r, C_1R = \frac{C_1C_2}{2} = \frac{r}{2}$$

Using Pythagoras theorem in ΔC_1PR ,

$$C_1P^2 = PR^2 + C_1R^2$$

Using Pythagoras theorem in ΔC_1PR ,

$$C_1P^2 = PR^2 + C_1R^2$$

$$\Rightarrow PR = \sqrt{r^2 - \left(\frac{r}{2}\right)^2}$$

$$\Rightarrow PR = \frac{\sqrt{3}r}{2}$$

$$\therefore PQ = 2PR = 2 \times \frac{\sqrt{3}r}{2} = \sqrt{3}r$$

Q9 (3) - Angle Between Circles and Radical Axis

The given equations of the circle are

$$x^2 + y^2 = a^2 \dots(i)$$

$$(x - c)^2 + y^2 = a^2 \dots(ii)$$

$$x^2 + (y - b)^2 = a^2 \dots(iii)$$

From Eqs. (i) and (ii) of radical axis, we get

$$c^2 - 2xc = 0$$

$$\Rightarrow x = \frac{c}{2}$$

From Eqs. (iii) and (i) of radical axis, we get

$$b^2 - 2by = 0$$

$$\Rightarrow y = \frac{b}{2}$$

So, the coordinates of centre $\left(\frac{c}{2}, \frac{b}{2}\right)$.

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Q10 (2) - Angle Between Circles and Radical Axis

The given equations of the circles are

$$S_1 \equiv x^2 + y^2 + x - y + 2 = 0$$

$$\text{and } S_2 \equiv x^2 + y^2 - \frac{4}{3}x - 4 = 0$$

The radical axis of two circles $S_1 = 0$ and $S_2 = 0$ is given by

$$S_1 - S_2 = 0$$

$$\Rightarrow (x - y + 2) - \left(\frac{4}{3}x - 4\right) = 0$$

$$\Rightarrow x + \frac{4}{3}x - y + 2 + 4 = 0$$

$$\Rightarrow 7x - 3y + 18 = 0$$

Q11 (3) - Angle Between Circles and Radical Axis

If two circles touch each other orthogonally, then

$$2(g_1g_2 + f_1f_2) = c_1 + c_2$$

Here, (g_1, f_1) is $(3, 4)$

and (g_2, f_2) is $(7, 8)$

$$\text{Then, } 2(21 + 32) = c_1 + c_2$$

$$\Rightarrow c_1 + c_2 = 106 \dots (i)$$

Since, radius of both circles are equal.

$$\text{So, } r_1 = \sqrt{g_1^2 + f_1^2 - c_1}$$

$$\text{and } r_2 = \sqrt{g_2^2 + f_2^2 - c_2}$$

$$r_1 = r_2$$

$$\Rightarrow r_1^2 = r_2^2$$

$$\Rightarrow -C_1 + C_2 = 88 \dots (ii)$$

On solving Eqs. (i) and (ii), we get

$$c_1 = 9 \text{ and } c_2 = 97$$

$$\text{Hence, } r_1 = \sqrt{9 + 16 - 9} \Rightarrow r_1 = 4 \text{ units}$$

Q12 (3) - Angle Between Circles and Radical Axis

Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0 \dots (i)$

It cuts the circle $x^2 + y^2 = 4$ orthogonally.

$$\text{Then, } 2(g_1g_2 + f_1f_2) = c_1 + c_2$$

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$$\Rightarrow 2(g \cdot 0 + f \cdot 0) = c - 4 \Rightarrow c = 4$$

and it passes through (1,2).

$$\therefore 1 + 4 + 2g + 4f + 4 = 0$$

$$2g + 4f + 9 = 0$$

So, locus of $(-g, -f)$ is

$$-2x - 4y + 9 = 0$$

$$\Rightarrow 2x + 4y - 9 = 0$$

Q13 (2) - Location of Circles

The given equations of the circles are

$$x^2 + y^2 = 4 \dots(i)$$

$$x^2 + y^2 - 6x - 8y - 24 = 0 \dots(ii)$$

Centres and radii of these circles are

$$C_1(0, 0), C_2 = (3, 4) \text{ and } r_1 = 2$$

$$r_2 = \sqrt{9 + 16 + 24} = 7$$

$$C_1C_2 = \sqrt{9 + 16} = 5$$

$$r_2 - r_1 = 5$$

$$|C_1C_2| = |r_1 - r_2|$$

$\therefore$ There is one common tangent.

Q14 (4) - Location of Circles

The given equations of the circles are

$$x^2 + y^2 + 2hx - (r_1^2 - h^2) = 0 \dots(i)$$

$$x^2 + y^2 + 2kx - (r_2^2 - k^2) = 0 \dots(ii)$$

The centres and radii of the circles are

$$C_1(-h, 0) \text{ and } (-k, 0)$$

$$R_1 = \sqrt{h^2 + r_1^2 - h^2} = r_1$$

$$R_2 = \sqrt{k^2 + r_2^2 - k^2} = r_2$$

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Since, they touch externally

$$|C_1 C_2| = R_1 + R_2$$

$$\sqrt{h^2 + k^2} = r_1 + r_2$$

$$\Rightarrow h^2 + k^2 = (r_1 + r_2)^2$$

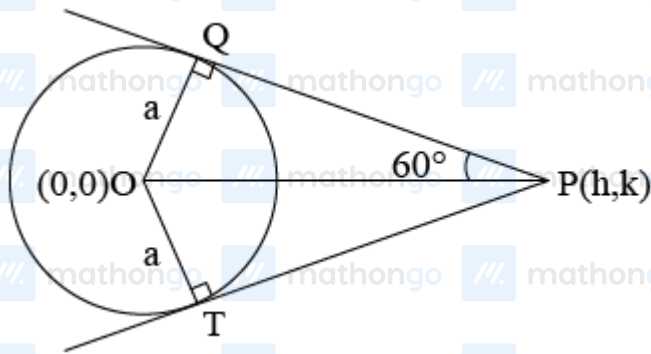
$$\therefore h^2 + k^2 = r_1^2 + r_2^2 + 2r_1 r_2$$

Q15 (1) - Location of Circles

If a circle S_1 bisects the circumference of another circle S_2 , then it will be the diameter of circle S_2 . Then, their radial axis passes through the centre of S_1 .

Q16 (2) - Location of Circles

The given equation of the circle is $x^2 + y^2 = a^2$



where, $O(0,0)$ is centre of the circle and a is the radius of the circle. Let the coordinates of point P be (h, k)

$$OP = \sqrt{(h-0)^2 + (k-0)^2} = \sqrt{h^2 + k^2}$$

In $\triangle POQ$, $\frac{a}{\sqrt{h^2 + k^2}} = \sin 60^\circ$

$$\Rightarrow \frac{a}{\sqrt{h^2 + k^2}} = \frac{\sqrt{3}}{2} \Rightarrow \frac{a^2}{h^2 + k^2} = \frac{3}{4}$$

$$\Rightarrow 3h^2 + 3k^2 = 4a^2$$

On taking locus of (h, k) , $3x^2 + 3y^2 = 4a^2$

Q17 (3) - Location of Circles

Here, $m_{PQ} = -\frac{4}{3}$.

$\therefore$ Coordinates of P , are



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$$(0 \pm 3 \cos \theta, 0 \pm 3 \sin \theta) \dots(i)$$

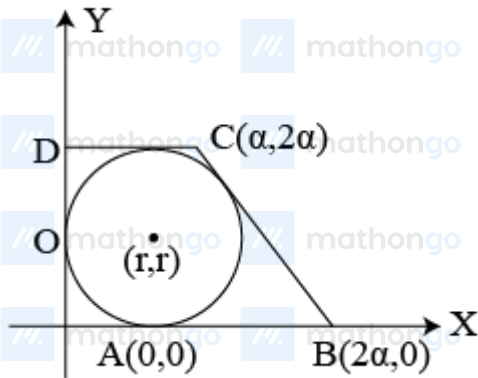
$$\text{Here, } \tan \theta = -\frac{4}{3} \Rightarrow \sin \theta = \frac{4}{5}$$

$$\text{and } \cos \theta = -\frac{3}{5}$$

$$\text{Thus, centre coordinates are } \left(\mp \frac{9}{5}, \pm \frac{12}{5} \right)$$

Q18 (2) - Location of Circles

Let $CD = \alpha$. Then, $AB = 2\alpha$



The equation of BC is

$$y = -\frac{2r}{\alpha}(x - 2\alpha)$$

$$\text{or } 2x + \alpha y - 4r\alpha = 0$$

$$\text{It touches the circle } (x - r)^2 + (y - r)^2 = r^2$$

$$\therefore \left| \frac{2r^2 + \alpha r - 4\alpha r}{\sqrt{4r^2 + \alpha^2}} \right| = r$$

$$\Rightarrow (2r^2 - 3\alpha r)^2 = r^2 (\alpha^2 + 4r^2)$$

$$\Rightarrow (2r - 3\alpha)^2 = \alpha^2 + 4r^2$$

$$\Rightarrow -12r\alpha + 9\alpha^2 = \alpha^2 \Rightarrow \alpha = \frac{3r}{2}$$

Now, area of quadrilateral $ABCD = 18$

$$\Rightarrow \frac{1}{2} \times (AB + CD) \times 2r = 18$$

$$\Rightarrow 3\alpha r = 18 \Rightarrow \alpha r = 6$$

$$\Rightarrow 3r^2 = 12 \quad \left[\because \alpha = \frac{3r}{2} \right]$$

$$\therefore r = 2 \text{ units}$$

Q19 (1) - Chord of Contact, Chord with Given Mid Point & Pair of tangents

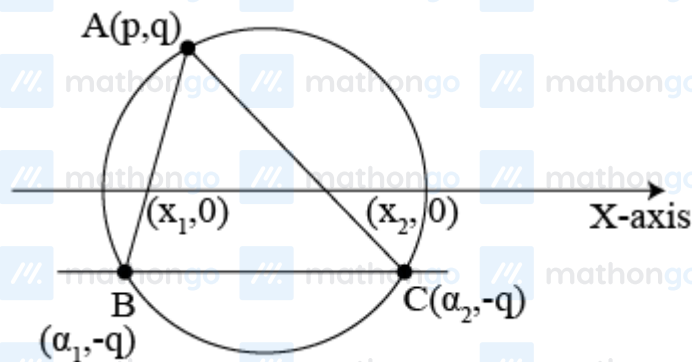
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$$SS_1 = T^2$$

$$\Rightarrow \left(x^2 + y^2 - \frac{1}{4}\right) \left(\frac{3}{4}\right) = \left(y - \frac{1}{4}\right)^2$$

Since, $y = -q$ intersect $x^2 + y^2 = px + qy$ at two real and distinct points.



$$\Rightarrow x^2 + q^2 = px - q^2$$

$$\Rightarrow x^2 - px + 2q^2 = 0$$

Here, $D > 0 \Rightarrow p^2 - 8q^2 > 0$

or $p^2 > 8q^2$

Let $P(x_1, y_1)$ be a point on the circle $x^2 + y^2 = a^2$. Then,

$$x_1^2 + y_1^2 = a^2 \dots(i)$$

The equation of the chord of contact of tangents drawn from (x_1, y_1) to the $x^2 + y^2 = b^2$ is

$$xx_1 + yy_1 = b^2 \dots(\text{ii})$$

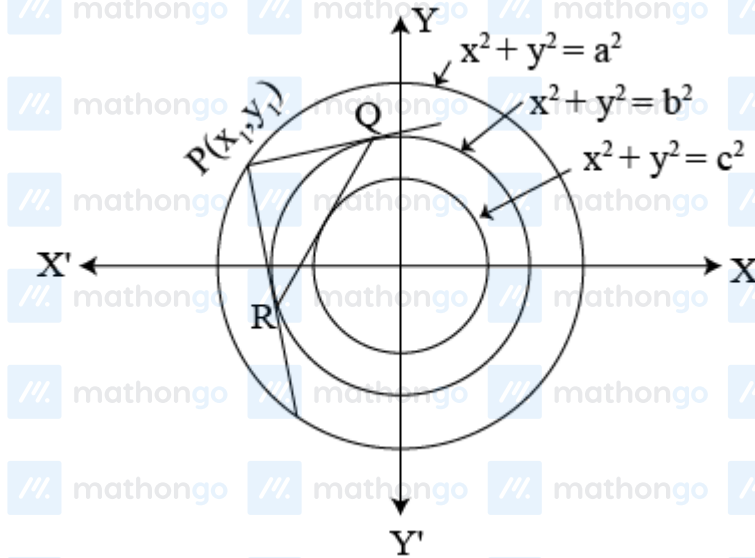
This touches the circle

$$x^2 + y^2 = c^2$$

$\therefore$ Length of the perpendicular from (0,0) to line (ii) is equal to the radius of $x^2 + y^2 = c^2$

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$$\Rightarrow \left| \frac{0 \times x_1 + 0 \times y_1 - b^2}{\sqrt{x_1^2 + y_1^2}} \right| = c$$

$$\Rightarrow \frac{b^2}{\sqrt{a^2}} = c$$

$$\Rightarrow b^2 = ac$$

$\therefore a, b, c$ are in GP.

Q22 (1) - Chord of Contact, Chord with Given Mid Point & Pair of tangents

Locus of mid-point (h, k) is $S_1 = T$

$$\Rightarrow h^2 + k^2 - 9 = xh + yk - 9$$

$$\text{or } xh + yk - (h^2 + k^2) = 0 \dots (i)$$

Also, let the point on $4x - 5y = 0$ be $\left(\lambda, \frac{4\lambda - 20}{5}\right)$

$\therefore$ Chord of contact of tangents is

$$x\lambda + y \cdot \left(\frac{4\lambda - 20}{5}\right) - 9 = 0 \dots (ii)$$

Since, Eqs. (i) and (ii) is same

$$\therefore \frac{h}{\lambda} = \frac{\frac{k}{4\lambda - 20}}{\frac{1}{5}} = \frac{-(h^2 + k^2)}{-9}$$

$$(4\lambda - 20)h = 5\lambda k$$

$$\text{or } \lambda = \frac{20h}{4h - 5k}$$

$$\text{and } \frac{h}{\lambda} = \frac{h^2 + k^2}{9}$$

$$\frac{9h \cdot (4h - 5k)}{20h} = h^2 + k^2 \text{ [using Eq.(iii)]}$$

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or locus is

$$9(4x - 5y) = 20(x^2 + y^2)$$

$$\text{or } 20(x^2 + y^2) - 36x + 45y = 0$$

Q23 (1) - Chord of Contact, Chord with Given Mid Point & Pair of tangents

The given equation of the circle is

$$x^2 + y^2 - 2x + 4y + 3 = 0$$

having centre (1,-2) and radius = $\sqrt{2}$ Equation of any line through (6,-5) is

$$y + 5 = m(x - 6)$$

$$\Rightarrow mx - y - 6m - 5 = 0$$

If it touches the circle, then length of perpendicular from centre

(1,-2) to the circle is equal to the radius $\sqrt{2}$.

$$\therefore \frac{m+2-6m-5}{\sqrt{m^2+1}} = \pm\sqrt{2}$$

$$\frac{-5m-3}{\sqrt{m^2+1}} = \pm\sqrt{2}$$

$$\Rightarrow (-5m-3)^2 = 2(m^2+1)$$

$$\Rightarrow 25m^2 + 9 + 30m = 2m^2 + 2$$

$$23m^2 + 23m + 7 = 0$$

$$\Rightarrow 23m(m+1) + 7(m+1) = 0$$

$$\Rightarrow m = -\frac{7}{23}, -1$$

So, equations of tangents are, if $m = -\frac{7}{23}$

$$y + 5 = -\frac{7}{23}(x - 6)$$

$$\Rightarrow 23y + 115 = -7x + 42$$

$$\Rightarrow 7x + 23y + 73 = 0$$

$$\text{If } m = -1, y + 5 = -1(x - 6)$$

$$\Rightarrow y + 5 = -x + 6 \Rightarrow x + y - 1 = 0$$

So, pair of tangent is

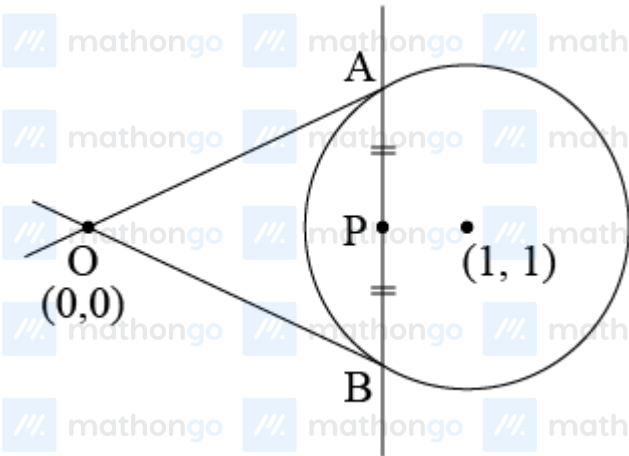
$$(7x + 23y + 73)(x + y - 1) = 0$$

$$7x^2 + 23y^2 + 30xy + 66x + 50y - 73 = 0$$

Q24 (1) - Chord of Contact, Chord with Given Mid Point & Pair of tangents

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Equation of AB is $S_1 = T$



$$\therefore h^2 + k^2 - 2h - 2k = xh + yk - (x + h) - (y + k)$$

$$\text{Or } h^2 + k^2 - h - k = xh + yk - x - y$$

$$\text{Or } 1 = \frac{x(h-1)+y(k-1)}{h^2+k^2-h-k} \dots(i)$$

Now, product of pair of tangents OA and OB is

$$x^2 + y^2 - 2x \cdot \left[\frac{x(h-1)+y(k-1)}{h^2+k^2-h-k} \right] - 2y \left[\frac{x(h-1)+y(k-1)}{h^2+k^2-h-k} \right] = 0$$

Since, OA and OB are equally inclined to axes.

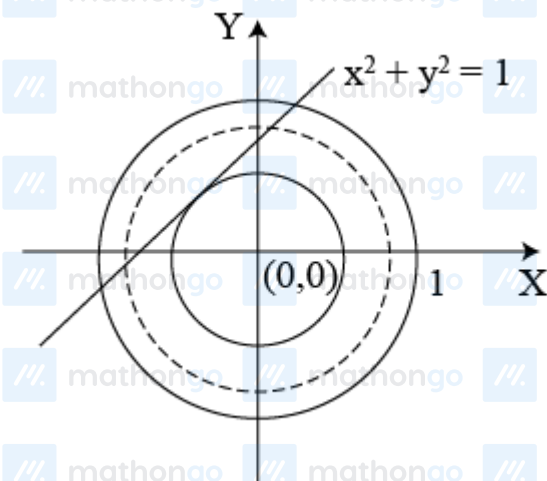
$$\therefore m_1 + m_2 = 0 \Rightarrow -2(k-1) - 2(h-1) = 0$$

$$\text{or } h - 1 + k - 1 = 0$$

$$\text{Or } x + y = 2$$

Q25 (3) - Chord of Contact, Chord with Given Mid Point & Pair of tangents

Radius of circle are r_1, r_2 and line $y = x + 1$



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Perpendicular from (0,0) on line $y = x + 1 \Rightarrow \frac{1}{\sqrt{2}}$ Now, $r_1 > \frac{1}{\sqrt{2}}$ but $r_1 = 1 - 2d$

Hence, $1 - 2d > \frac{1}{\sqrt{2}}; \frac{\sqrt{2}-1}{\sqrt{2}} > 2d$

$$d < \frac{\sqrt{2}-1}{2\sqrt{2}} \Rightarrow d \in \left(0, \frac{\sqrt{2}-1}{2\sqrt{2}}\right)$$

Q26 (3) - Pole and Polar

Let pole be (x_1, y_1)

$\therefore$ Equation of polar, is

$$2x(x_1) + 2y(y_1) - \frac{3}{2}(x + x_1) + \frac{5}{2}(y + y_1) - 7 = 0$$

$$\Rightarrow 2xx_1 + 2yy_1 - \frac{3}{2}x - \frac{3}{2}x_1 + \frac{5}{2}y + \frac{5}{2}y_1 - 7 = 0$$

$$\Rightarrow x\left(2x_1 - \frac{3}{2}\right) + y\left(2y_1 + \frac{5}{2}\right) - \frac{3}{2}x_1 + \frac{5}{2}y_1 - 7 = 0$$

is same as, $9x + y - 28 = 0$

$$\Rightarrow \frac{2x_1 - 3/2}{9} = \frac{2y_1 + 5/2}{1} = \frac{(-3/2)x_1 + (5/2)y_1 - 7}{-28}$$

On solving $x_1 = 3, y_1 = -1$

Q27 (1) - Pole and Polar

Let pole be (h, k) .

Then, equation of polar is $xh + yk = 5$ same as

$$\therefore x + 2y = 1$$

$$\frac{h}{1} = \frac{k}{2} = \frac{5}{1}$$

$$\Rightarrow h = 5, k = 10$$

Thus, pole is (5,10)

Q28 (2) - Pole and Polar

Let (x_1, y_1) be the pole of the given straight line w.r.t. $x^2 + y^2 = c^2$. Then, the equation of the polar, i.e. the straight line is

$$xx_1 + yy_1 = c^2 \dots(i)$$

It is given that, (x_1, y_1) lies on

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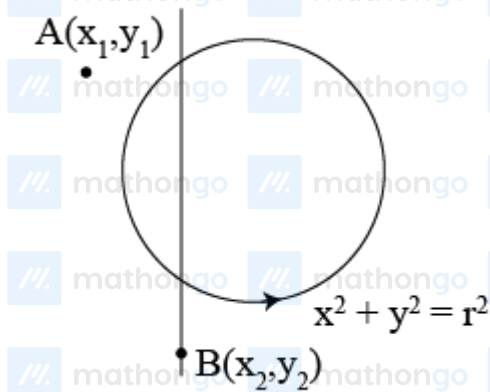
$$x^2 + y^2 = 9c^2$$

$$\therefore x_1^2 + y_1^2 = 9c^2$$

Now, length of the perpendicular from the centre $O(0, 0)$ of the circle $x^2 + y^2 = \frac{c^2}{9}$ to the line (i) is equal to the radius of the circle. Hence, the straight line touches the circle $x^2 + y^2 = \frac{c^2}{9}$.

Q29 (2) - Pole and Polar

Here, $A(x_1, y_1)$ and $B(x_2, y_2)$ are conjugate points w.r.t. $x^2 + y^2 = r^2$



$\therefore$ Chord of contact for $A(x_1, y_1)$ is $T = 0$

$\Rightarrow xx_1 + yy_1 = r^2$ passes through $B(x_2, y_2)$

$\Rightarrow x_1x_2 + y_1y_2 = r^2$

Also, $l_1^2 = x_1^2 + y_1^2 - r^2$

and $l_2^2 = x_2^2 + y_2^2 - r^2$

$\therefore AB^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$

$= x_1^2 + y_1^2 + x_2^2 + y_2^2 - 2(x_1x_2 + y_1y_2)$

$= l_1^2 + r^2 + l_2^2 + r^2 - 2r^2 = l_1^2 + l_2^2$

Q30 (1) - Pole and Polar

Let from $P(x_1, y_1)$ to the circle $x^2 + y^2 = a^2$ is a chord of contact of tangents.

Since, it passes through $(k, 0)$, therefore

$x_1 \cdot k + y_1 \cdot 0 = a^2$

$\Rightarrow x_1k = a^2$

$\therefore$ Locus of $P(x_1, y_1)$, polar is

$kx = a^2$

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